

A Trend-Filtered Volume Profile Mean-Reversion Strategy: Design, Backtest, and the Primacy of Risk-Reward over Win Rate

A systematic study on QQQ intraday data

Abstract

This study documents the design and systematic backtest of a trading strategy built around the Volume Profile — a representation of traded volume distributed across price levels. Five progressive variants were evaluated on 60 days of 5-minute QQQ data. The central finding is that the risk-reward ratio, rather than the win rate, determines profitability: the variant winning only one trade in three outperformed the variant winning nearly two in three, owing to tightly placed stops anchored to low-volume nodes. The best configuration returned +4.73% with a maximum drawdown of 1.72%, though results are conditioned on a trending sample and should be interpreted with the stated limitations in mind.

1. The Volume Profile

A conventional price chart represents price as a function of time. The Volume Profile inverts this relationship, aggregating traded volume by price level to reveal where market participants transacted most heavily. Three reference levels follow from this construction and form the basis of the strategy:

- **Point of Control (POC)** — the price level of greatest traded volume, interpreted as the market's fair-value consensus and a magnet to which price tends to return.
- **Value Area (VA)** — the contiguous range containing seventy percent of traded volume, bounded above by the VAH and below by the VAL.
- **Low Volume Node (LVN)** — a price level of minimal traded volume, representing a zone the market transited rapidly and seldom accepted.

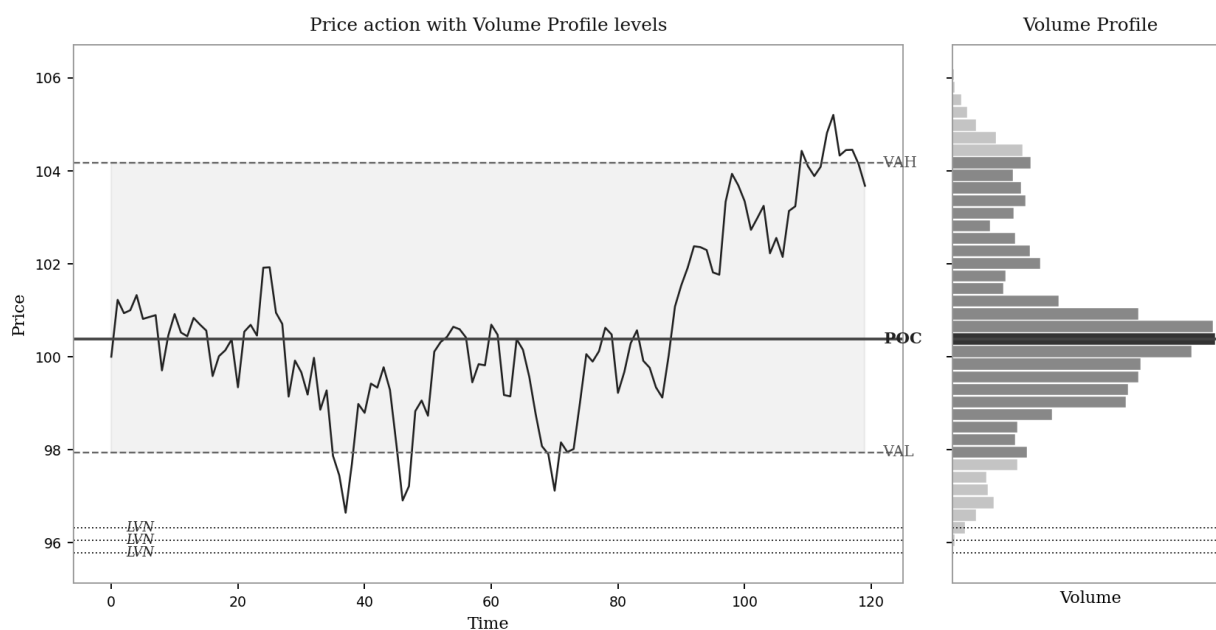


Figure 1. Schematic of a Volume Profile. The histogram (right) shows volume per price level; the POC marks its peak, the shaded band the Value Area between VAH and VAL, and the dotted lines the low-volume nodes used for stop placement.

2. Trading Hypothesis

The strategy implements a trend-filtered reversion to value. When price drifts to the edge of the value area while the prevailing trend remains intact, it is hypothesised to revert toward the POC. Trades are taken only in the direction of the trend, entering on pullbacks to value rather than opposing momentum. Pure mean reversion, by contrast, trades against the trend and is shown below to be unprofitable; the addition of an EMA(3/9) trend filter is the single change that renders the same logic profitable.

3. Trade Rules

For a long position the procedure is as follows. First, the trend condition requires EMA(3) above EMA(9). Second, entry is triggered when price retraces to the prior period's VAL. Third, the protective stop is placed immediately beyond the nearest low-volume node below the VAL; a breach of this zone invalidates the premise of the trade. Fourth, the profit target is the POC, or alternatively the VAH for a larger but less frequently achieved objective. Positions unresolved at the close of the period are exited at market. Short positions mirror this logic about the VAH.

The low-volume node merits emphasis as the logical stop location. Because such nodes mark prices the market rejects, a return through one against the position signals structural failure. Their proximity to the entry keeps the stop tight, which in turn elevates the achievable risk-reward ratio — the mechanism underlying the study's principal result.

To avoid look-ahead bias, each period is traded exclusively on the preceding period's profile: the daily variant trades today on yesterday's levels, the weekly variant trades this week on last week's.

4. Results

Five variants were evaluated on identical data (QQQ, 60 days, 5-minute bars). Table 1 reports the outcomes.

Variant	Description	Trades	Win Rate	Return	Avg. R:R
v1	Pure mean reversion	238	58.0%	−8.59%	—
v2	Trend filter, POC target	164	64.6%	+0.75%	—
v3	Trend filter, extended target	99	34.3%	−7.48%	—
v5	LVN stops, R:R filter	87	33.3%	+4.66%	6.51

Table 1. Performance of the five strategy variants. The best variant (v5) achieved a maximum drawdown of 1.72%.

5. Discussion

The results invert a common intuition. The variant with the lowest win rate (v5, 33.3%) produced the highest return, while the highest win rate (v1, 58.0%) produced the largest loss. The reconciliation lies in the asymmetry of outcomes: v5's low-volume-node stops yield an average reward-to-risk ratio of 6.5 to 1, so its infrequent winners substantially exceed its frequent small losers. A favourable risk-reward ratio thus dominates a high win rate as a determinant of profitability.

A comparison of timeframes found the daily profile superior to the weekly (+4.66% against +0.68%) with a smaller drawdown. The weekly variant exhibited a higher theoretical risk-reward (12.58) but was undermined by an insufficient sample of thirteen weeks, preventing its statistical edge from being realised.

6. Limitations and Practical Considerations

Several caveats temper these findings. The sixty-day window is short and the results are not statistically robust. The test period was strongly directional, and the strategy requires validation in range-bound conditions. Transaction costs and slippage are not modelled, and freely available intraday data constrains both historical depth and volume granularity.

A behavioural consideration is equally material. A win rate near one-third entails extended sequences of losing trades preceding each winner. Although the expectancy is positive, sustaining such a sequence demands discipline that not all traders possess, and this should be weighed before any live deployment.

This document summarises a personal research project for educational purposes. It does not constitute financial advice. Backtested performance does not guarantee future results.